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SRE

Software Engineering: Theories, methods and tools to make professional software.

Theory → Real-world Paradigm.

Methods → ways to exercise the theory.

Entity Relationship module.

OO Development.

Tools → Editors ; languages.

System Engineering

- Process Engineering (Business process).
- Hardware Engineering
- Software Engineering

Software Engineering

- SRE
- Design and Architecture
- Programming
- Testing and validation.

• Methods and tools to analyze, design, code and test the Software.

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NOTE: Asal maze 25 ke baad.

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Software Engineering Requirement

Identify Analyze: validate specify.

- Methods and tools to identify, analyze, validate and specify the requirements.

- **Requirements** → what system will do not how it should do it.
- Based on needs not wants.

External Behaviour → I / O

System: systematic approach to perform specific tasks.

System Requirements: The process, software and hardware requirements

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- **software Requirements may be:**
  - Abstract statements of services / constraints
  - Detailed Mathematical functions.

NOTE: No one is wrong, every person has a Perspective.

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• SRE → Not about I/O, it's about whether something is useful and needed or not.

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• **Software Requirements may be:**

- Part of bid of contract. (which company will build the product)
- The contract: (Terms and condition)
- Technical document. (for designers etc).

(for Requirements)

Mathematical languages:

1. Z specification language
2. VDM (Vienna Development Model)

IEEE: Condition or capability that must be possessed by a system, to satisfy a contract, standard, specification or other documents.

(formally imposed)

(Law from gov OR regulations from regulatory body).

Sources of Requirements:

• **Stackholders:** People affected by the system.

• **Documents:** Describe the I/O from the business processes.

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- **Existing System:** Previous system for understanding.
- **Domain / Business Area:** Specific domain constraints.

Levels of SR: Different levels of detail. (What vs How).

Importance of SR: "What to build".

- Ruins the whole system if done incorrectly.

4. **Functional Requirements:** What the system does.
   - Reaction to particular inputs.
   - Behaviour in particular situations.
   - Sequencing (Dependency) and parallelism (independent) are also captured by FR.
   - Exception handling.
   - FR should be complete & consistent.
     (I/O)

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2. Non-Functional Requirements:

Relates to the whole system.

- Constraints on timing, security, performance, accuracy, reliability, maintainability, the development process, standards etc.
- Emergent behaviour of system.
- More critical than FR.
- Built into framework.

NFR arise From:

Budget constraints, policies, interoperability with other software and hardware, safety regulations or legislation etc.

NFR

- Product Requirements
- Organizational Requirements
- external Requirements.

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4. : Product Requirement

usability Efficiency Reliability Portability.

Performance space

- **Product Requirement:** Qualities of the software product.

i. **Usability** → Easy to learn and understand.

ii. **Efficiency** → How well the system uses resources.

- **Performance:** Evaluated based on number of lines of code and how much data is loaded in memory for execution of code. How fast the system is.
- **Space:** storage and memory needed.

iii. **Reliability** → correctness and consistency of output also availability of system.

- Mean time to failure.
- Mean time to Recover.

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iv. **Portability** → system should work the same on different environments and operating systems.
→ Platform dependent system.

organizational Requirements
↓ Standards ↓ Implementation ↓ Delivery.

- **organizational Requirements:** Rules of an organization (Medical standards).
  Includes: • Developers • stackholders.

i- **Standards** → Set of rules that must be followed.

ii- **Implementation** → [→ Making the system live.
- **Phase cutover** (implementation in parts, replacing only a part that is older)
- **Parallel Run** → New and old system run in parallel, the outputs are compared for reliability.
- **Direct Replacement** → Replacing the entire old system. (For clear requirements)

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iii. **Delivery** → where and how the system will be delivered.

External Requirements

↓ interoperability ↓ Ethical ↓ legislative

↓ privacy ↓ safety.

- **External Requirements:** Rules from outside the organization.

i. **Interoperability** → working with other system. communication and exchange of data with another system.

ii. **Ethical** → Responsible and morally acceptable way of building software. (applies to whole community).

iii. **Legislative** → Requirements enforced by Law.

- **Privacy** → Protection of personal info.
- **safety** → Protection from harm.

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and revision).

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- Priority of one requirement over another is decided based on the alignment of the requirement with business goals.

Objective verification → Quantification of requirements. (From general goals to testable requirements).

Metrics → combination of measures.

Goals → unverifiable

NFR → verifiable.

Metrics for NFR's.

1. **Speed** → • Processed Transactions.
   - Response Time
   - Screen Refresh Time.
2. **Size** → • K Bytes (instructions).
   - Number of function points
3. **Ease of use** → • Training time.
   - Number of help frames. (less = Better).
4. **Reliability** → • Mean time to failure.
   - Probability of unavailability. (Recovery Time)
   - Rate of failure occurence
   - Availability.

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Task → event

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5. **Robustness** → · Time to restart after failure. (start of transaction)

· Percentage of events causing failure.

· Probability of data corruption on failure.

(higher Probability, less robustness) → To avoid we do backups, indexes and Transaction logs.

event logs.

↓

last event before failure.

6. **Portability** → · Percentage of target-dependent statements (The less better).

· Number of target systems.

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3. **Domain Requirements**: Requirements that come from application domain.
→ Can be Functional or Non-Functional.
→ Impose strict contraints on solutions.

4. **Inverse Requirements**: What the system shall not do
→ Indicate indecisive nature of customers.

5. **Design and implementation constraints**:
Development guidelines within which the designer must work.
→ limit design and implementations options.
→ Also impacts human resources.

Another view of Requirements:
- user requirements
- System contract requirements.

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Task → fully dressed usecase?
Pre case condition
Post case condition
normal flow condition
Buisness rationale.

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(only external)

- **User Requirements:** stated in natural language with help of forms and diagrams.

Management of changing requirements is done through traceability.

A requirement has to traceable in 3 directions.

1- Requirement to source traceability.
(links the req to its og stakeholder)

2- Requiremet to requirement traceability
(link b/w a req and other depending reqs)

3- Requirement to design traceability.
(links req with design module)

Fully Dressed usecase: A complete, detailed description of a usecase.

Pre condition: must happen before the use case is started.

Post condition: must happen after use case is completed.

Normal Flow: steps followed to complete a usecase.

Business Rationale: Reason why usecase is needed.
- How users use system to achieve a goal.

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McCall's Model: Divides software quality into 3 main categories.

1. **Product operation:** How well the software works.
2. **Product Revision:** How easily the software can be changed.
3. **Product Transition:** How easily the software can move to other environments

1-) **Correctness** → Does the system do what it's supposed to?

Reliability → Does it work consistently without failing? (99% system uptime)

Efficiency → Does it use time and resources well? (Response time = 2sec)

Integrity → Is the data and system protected? (0 unauthorized access)

Usability → Easy for users to learn and use? (9/10 complete task)

2-) **Maintainability** → How easy is it to fix bugs? (an hour to fix)

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Flexibility → How easy is it to modify or add features? (80% changes can be made without modifying core code).

Testability → How easy is it to test the software? (90% functions can be covered by tests)

3-) Portability → Can it run on other system? (Run on 3 different OS).

Reusability → Can parts of software be reused? (40% of components reused).

Interoperability → Can it work / connect with other systems? (4 external systems)

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- **System Contract Requirements:** sets system services and constraints.

→ Serves as basis of contract for implementation of the system.

→ More technical.

→ Should be complete (I/O, → normal flow, abnormal flow.) and consistent (no conflict), (→ usually between NFR.)

→ used by designers and developers as starting point for system design.

→ often described by natural language and other specification language (UML).

Requirements Problems: inconsistent and incomplete. Do not reflect the real needs of the customer.

→ Expensive to change once the requirements are agreed upon.

Problems with Natural Language:

- Lack of clarity.
- Requirements confusion.
- Requirements Amalgamation.
- Ambiguity in similar terms.
- overflexible.
- Impossible to modularize.

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Mo Tu We Th Fr Sa Su (HOW Part)

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- Processes involved in developing system requirements known as **Requirement Engineering**.

Input → Processing → output

Input:

- **Existing System information:** Requires info about two system.
  1. The system it's going to replace. (it's functionalities).
  2. The system it's going to interact with. (external NFR) (interopability).
- **Stakeholder needs:** what the stake-holder needs from the system to support their work.
- **Organizational standards:** Rules of an organization regarding system development practice, quality management etc.
- **Regulations:** External regulations such as health and safety regulations.
- **Domain information:** Specific application domains. (any constraints)

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Process:

- Requirement Engineering
  - Elicitation
  - Analysis & Negotiation
  - specification
  - validation

conflicting.

Output:

- **Agreed Requirements:** when stake-holders agree on certain requirements. These requirements are known as agreed requirements. (user).
- **System specification:** Detailed specification of a system.
- **System Models:** The requirements are supported by these models. usecase, dataflow etc.

(Not a sequential process)

RE Process: very critical process. (if this is wrong everything after will be wrong).

→ Tailored to needs.

We have two main tasks in RE process.

- **Problem Analysis:** software problem.
- **Product description:** what the system is and will do.

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1. **Requirement Elicitation:** Determining system requirements through consultation with stakeholders, from (Hardware, software, process documents) system documents, domain knowledge and market studies.

(identification, discovering and creation)

(done in light of business objective)

2. **Requirement Analysis & Negotiation:**

- Different activities usually performed simultaneously.

- Classification → Grouping and relating the requirements.
- Conflict Resolution → solving disagreements b/w different requirements of stakeholders.
- Prioritization → Deciding what task should be done first

→ Requirements are analyzed (classification part) from 3 perspectives.

- Business flow Perspective (kya) (uses flow charts) (or workflow) (order of events)
- Data flow perpective (kise) (Data produced or needed / required). (The data passed in the work flow). (uses data flow model)
- Role action perspective (kon) (who is doing what).

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3- Requirement Specification:

Building a tangible model in natural language and diagrams. Representation of requirements that can be assessed for correctness, complete-ness and consistency.

4- Requirement Validation:

Reviewing requirements model for consistency and completeness. Intended to detect problems in requirements document.

Requirements management: Not a part of RE process but happening throughout.

- version controlling
- Traceability.

→ Identify control and track requirements.

Requirements Document: Detailed descriptions of required system.

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• Model → Representation.

(evolutionary → vertically vertical)

• Types of prototyping system:

1. Evolutionary (change according to feedback) (evolve).
2. Throw away (feedback) (like front end).

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(web functional)

Can be done through:
1. Horizontal (one layer of architecture).
2. vertical (goes through all layers).

Actors → People involved in the execution of that process. Actors are identified by their roles rather than individually.

E.g. syster engineer, project manager etc.

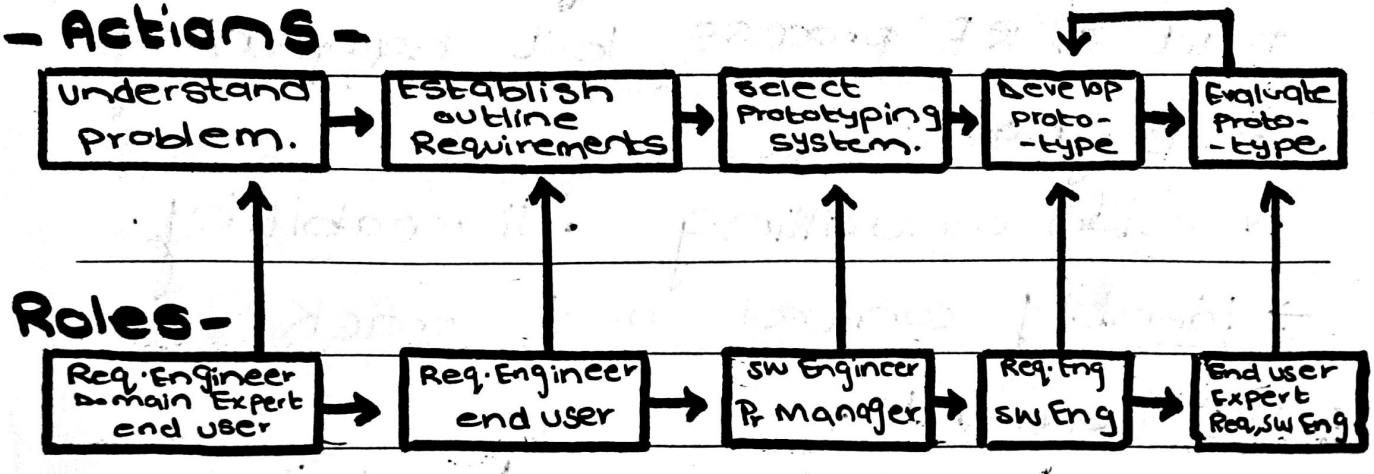
Actors in RE Process:

- People who are primarily interested in the problem to be solved (end user).
- People interested in the solution.
- Also external people.

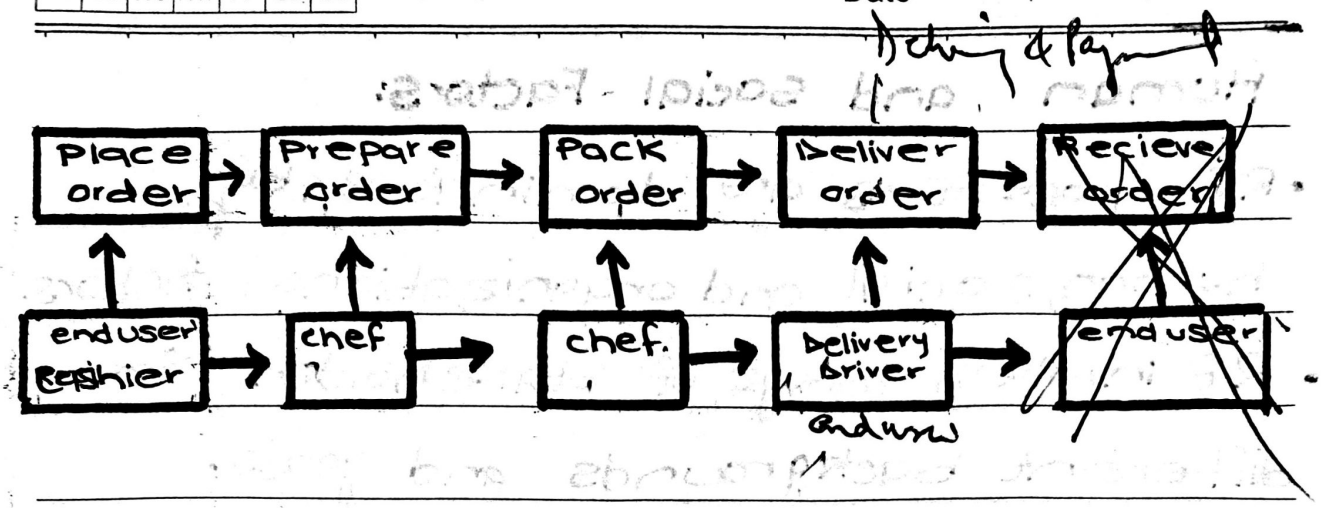
(Requirement to source traceability).

Role-action Diagram: Document the information needs of different people involved in the process.

E.g.1



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(for Eg 1)

Roles and Description:

- **Domain Expert:** Provides info about application domain.
- **End user:** using system after delivery.
- **Req Eng:** Elicitating & specifying req.
- **Project Manager:** Responsible for planning and estimating prototyping project.

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Human and social Factors:

- RE processes are dominated by human, social and organizational factors.
- RE involves range of stakeholders from different backgrounds and goals, some technical, some non-technical.
- Personal goals.
- Status of stakeholders.
- Political influence.

Social and cultural issues: